“Budget App” Prototype

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OVERVIEW

I will identify and solve the problems I face with managing my expenses and projecting a savings goal. I will be creating a prototype that helps the user track their income and expenses.

SPECIFICATIONS

The app will include expense tracking, budget setting and monitoring tools, and visual graphs for savings goals. The app will also include bill trackers, and notifications of upcoming bills. The app will sync with banking API’s to show the user’s balance, to make managing expenses even easier.

Milestones

1. ### Brainstorming

   I will identify the problem I have or might have when setting myself a budget and saving goal. a problem / solution space surrounding these problems

2. ### Lo-Fi Prototype

   Sketch initial concepts on paper, figure out what needs to be in the app and on each screen, and focus less on details.

3. ### Hi-Fi Prototype

   Develop detail into the wireframes, and add realistic dummy data wherever there should be data

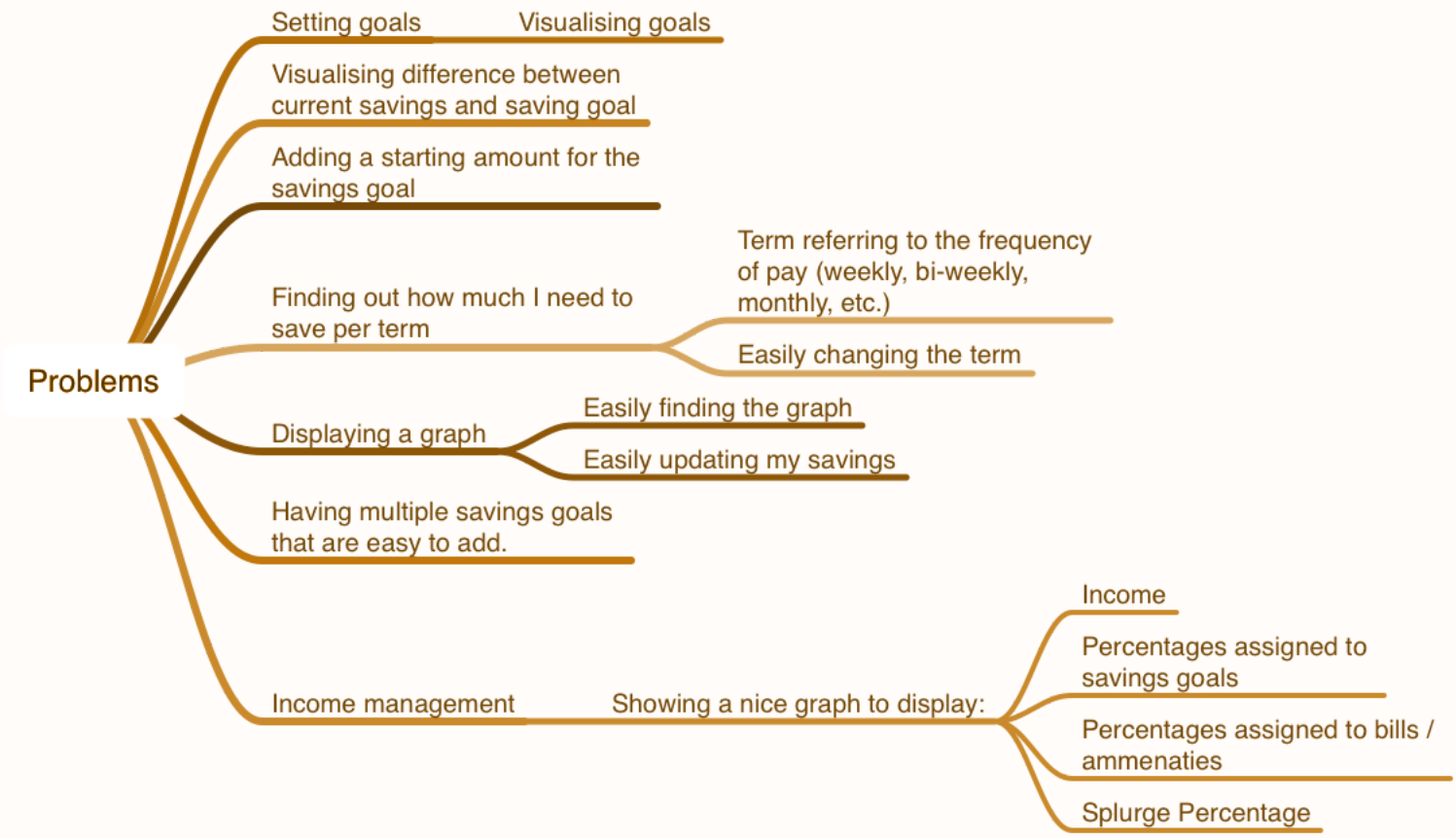
4. ### Hi-Fi WireFlow

   Prototype flows from screen to screen connected with buttons and interactions.

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Brainstorming



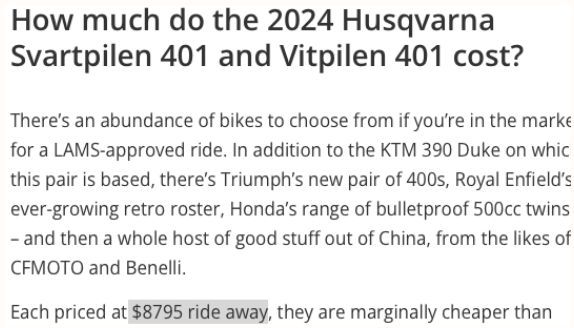
Issue Documentation

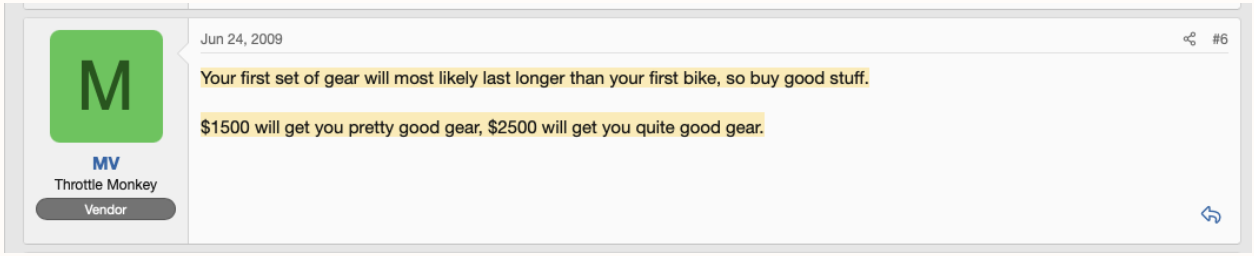
Setting Saving Goals

Issue: To set a saving goal, i want to easily be able to add a start point, end point, and saving interval. My current strategy using excel is inefficient to set up and use

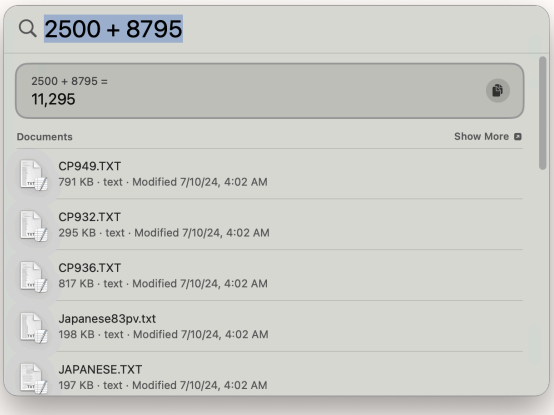
Step 1: Identify and add together each item involved in the total price

1. Search through google





2. Calculate approximate price of everything combined



Rounded up to $11,500 to be even, accounting for some smaller additional expenses.

Step 2: Open excel and add information for start date, as well as total price

		Values		
Name	Amount	Start Date		
Motorbike	$11,500.00	2-Aug-24		

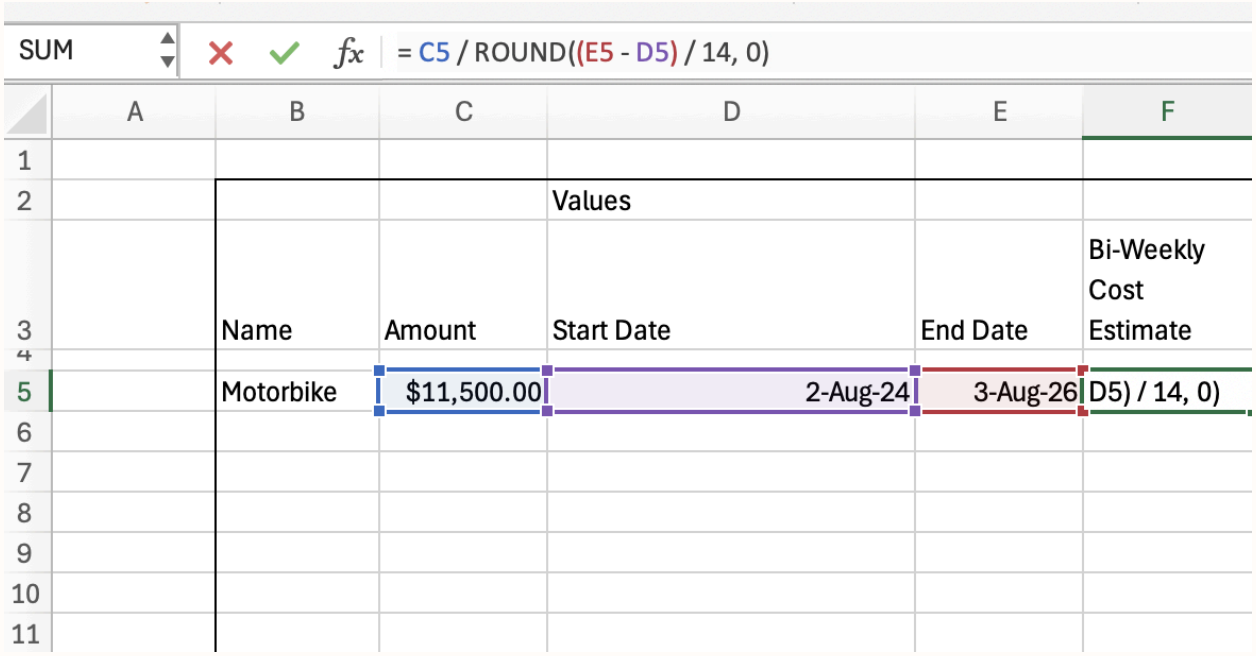
My next payment is the 2nd of august

Step 3: Identify and input my payment term, and the amount im comfortable saving from each payment

		Values		
Name	Amount	Start Date		Bi-Weekly Cost Estimate
Motorbike	$11,500.00	2-Aug-24		<$200

I usually spend more than 200 dollars for each fortnight I get paid, on things I don't need, so I can comfortably cut them out of my budget and instead save it towards the bike.

Step 4: add all the excel calculations and mathematical formulas



I'm not very accustomed to excel formulas, gotta give credit to chatGPT and google for being great tools for learning how to write them..

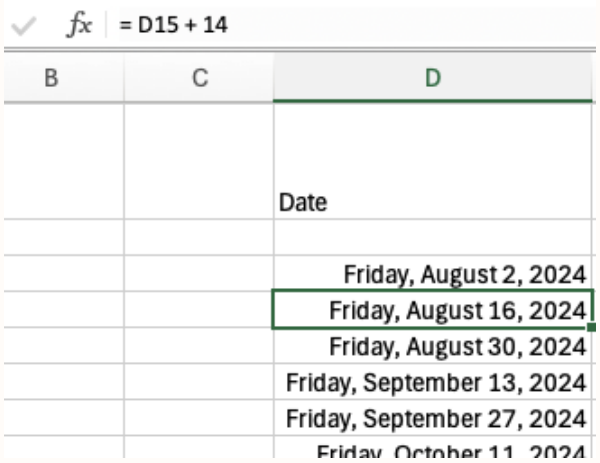
Step 5: Move end date back until my term cost aligns with my predetermined budget

		Values		
Name	Amount	Start Date	End Date	Bi-Weekly Cost
Motorbike	$11,500.00	2-Aug-24	2-Dec-26	$188.52

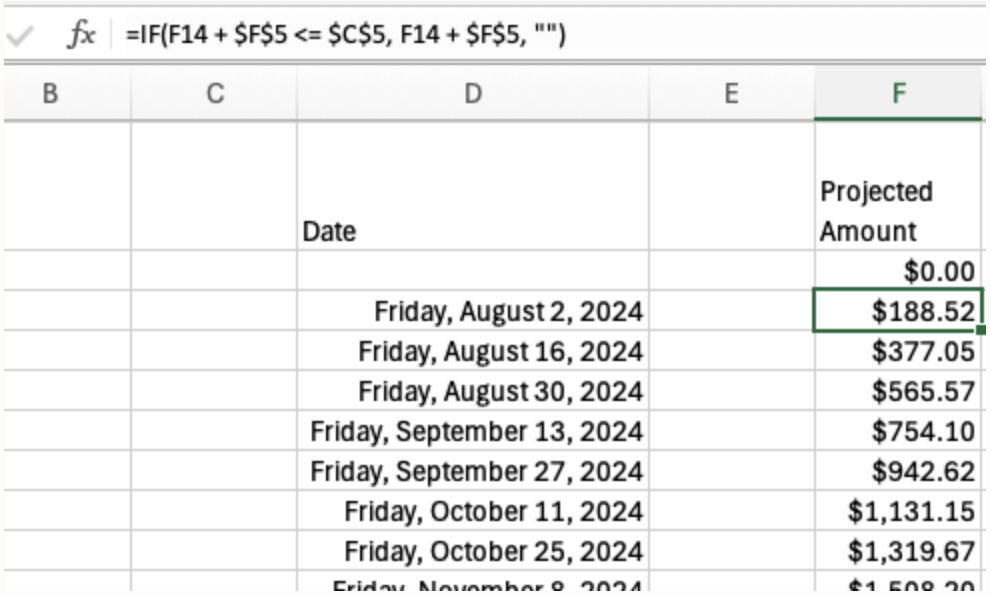
Since my current calculated savings are too high, I'll move the end date back a bit to fall below my target value.. This takes some trial and error

Step 6: Add forms of visualisation and term-based projected goals with inputs for my current savings amount

1. Date visualisation

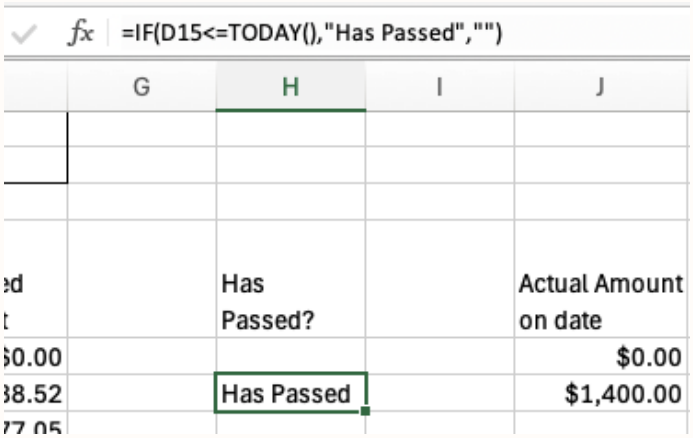


2. Projected amount which i can change if i get more to deposit

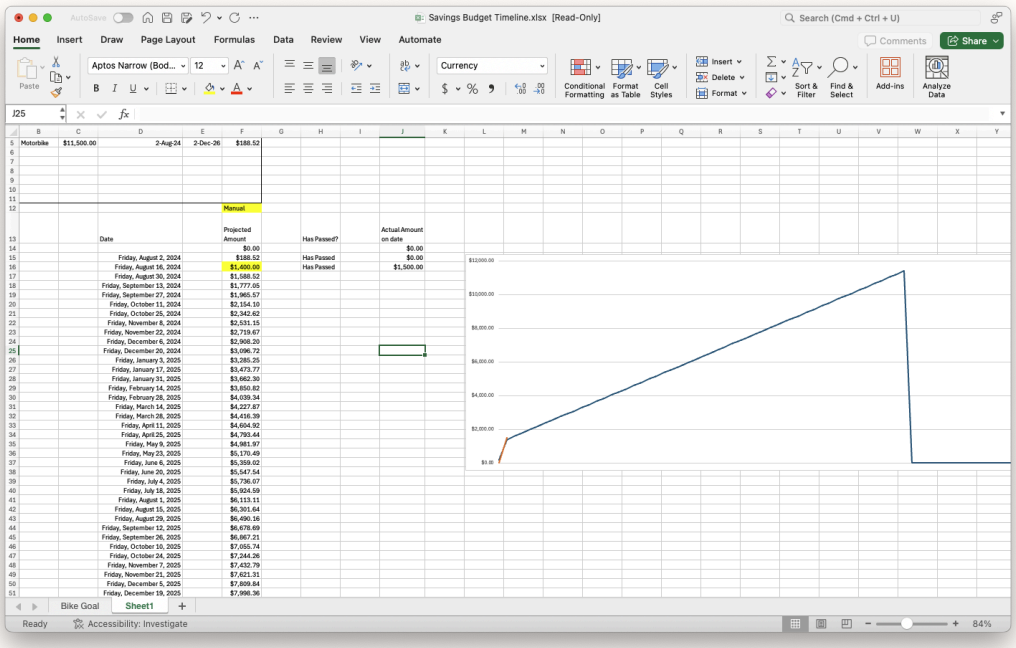


These formulas take a while to learn..

3. Automatic ‘has passed’ indications and input field for actual amount



4. A Graph



Updated for 18-Aug to include sample data - 1400 dollars to put into the account to start with, for example. The graph responds dynamically, so do the amounts per date.

What is the nature of the task I struggle to complete?

Money and Purchase Management

What goals do I have in mind when I carry out the task?

I want to save over a long period of time, to buy an item

What are my feelings before, during and after the task?

Before

- Excited about the thing i'm going to get

During

- Hating using excel
- Not loving the look of everything
- wishing it was easier to make inline edits

After

- Happy it's done
- Not wanting to do it again
- Still excited about getting whatever it is that i want, in whatever time frame
    - worried i might let myself spend the money on something else

What am I thinking before, during and after the task?

Before

- What is it that I want?
- Is there a time period I want to get this by?
- Weighing up my current money
    - How much do i earn
    - How much do i spend
    - How often do i get paid
        - How much can i comfortably afford to save per paycheck
    - How much money can i put aside as an initial deposit

During

- Why does excel have to be so difficult to use for me
- What even is a formula anymore?

- Checking my purchase history on my banking apps to see how much i spend on what

After

- Glad i got it done, there's a neat chart now
    - but i still need to save
    - and i don’t know if i can keep myself from spending the money i put aside

What contexts am I completing the task in?

Location

Usually at home, can’t imagine setting a saving goal anywhere else

Timeframe

Hoping for it to take maybe 5-15 minutes, realistically it takes over an hour (excel does not seem to offer much for beginners)

Actors

My bank account(s)

Excel

Google (gotta find the best price for what i’m looking for)

Artefacts

My phone

My computer

The item / thing i’m saving up for

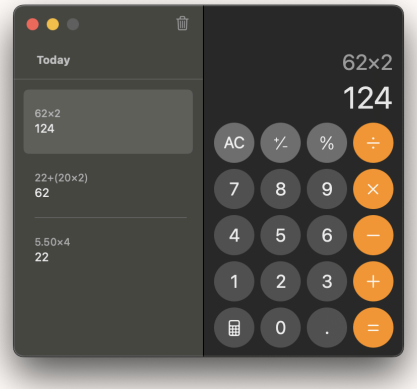
Logging and Organizing expenses.

Issue: I want to have an easy way to manage both my digital (card) spendings, and my physical (cash) spendings. My current strategy just involves doing some math every time i get paid to project my spendings

Step 1: Check how much i have to spend

I would check my westpac account after depositing money into savings, myki, etc.

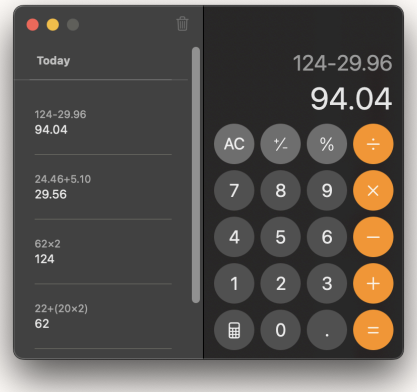
Step 2: Calculate how much i need for food and drink until i get paid



For each fortnite, 2 x 4 days a week at uni, one coffee ($5.50x4) a day.. Two dinners ($20x2) a week for the late classes.

Step 3: Add money to spendings account

1. *Log into my banking app, find my spendings balance. (after paying for myki, phone, etc)*
2. *Look in my wallet for cash if I have any, to put towards my budget.*
3. *Add both together, then subtract total from budgeted amount*



4. *Transfer money to spendings account*

Note that I am not currently logging expenses, I often forget about extra things i spend money on (snacks, lunch, etc) with no way of remembering to account for it.

Step 4: Repeat every fortnight

What is the nature of the task I struggle to complete?

Money and purchase management

What goals do I have in mind when I carry out the task?

I want to be able to find all my expenditures in an easy way, quicker and more efficient than just setting a budget throughout the week or spending and re-calculating as i go

What are my feelings before, during and after the task?

Before

- potentially a bit hesitant, because i know it will take a while to calculate, and it wont always be accurate

During

- Frustrated that i don't have as much as i want or thought i had
- uncomfortable with the amount of maths i'm doing, the less I do the better.

After

- Happy to at least have a budget now
- Regretful as i probably have to calculate everything again as i spend throughout the week

What am I thinking before, during and after the task?

Before

- Thinking about how long it might take
- I'm probably not going to enjoy it

During

- Wondering if there was some better way to do it

After

- Maybe i could go back and check if i can get myself something nice with spare money
- After spending money on an unexpected expense, I'm wondering if I have enough money for everything else, or if I have to dig into my savings..

What contexts am I completing the task in?

Location

My home

In the city

Shopping

At cafe's / Pubs / Bars / Diners

Timeframe

Usually 10-20 minutes

(longer than it should take)

Actors

The shopping centre

My banking apps

Calculator

Artefacts

Cash on hand

Credit cards

Food / Drink / groceries

Prototype Brainstorming

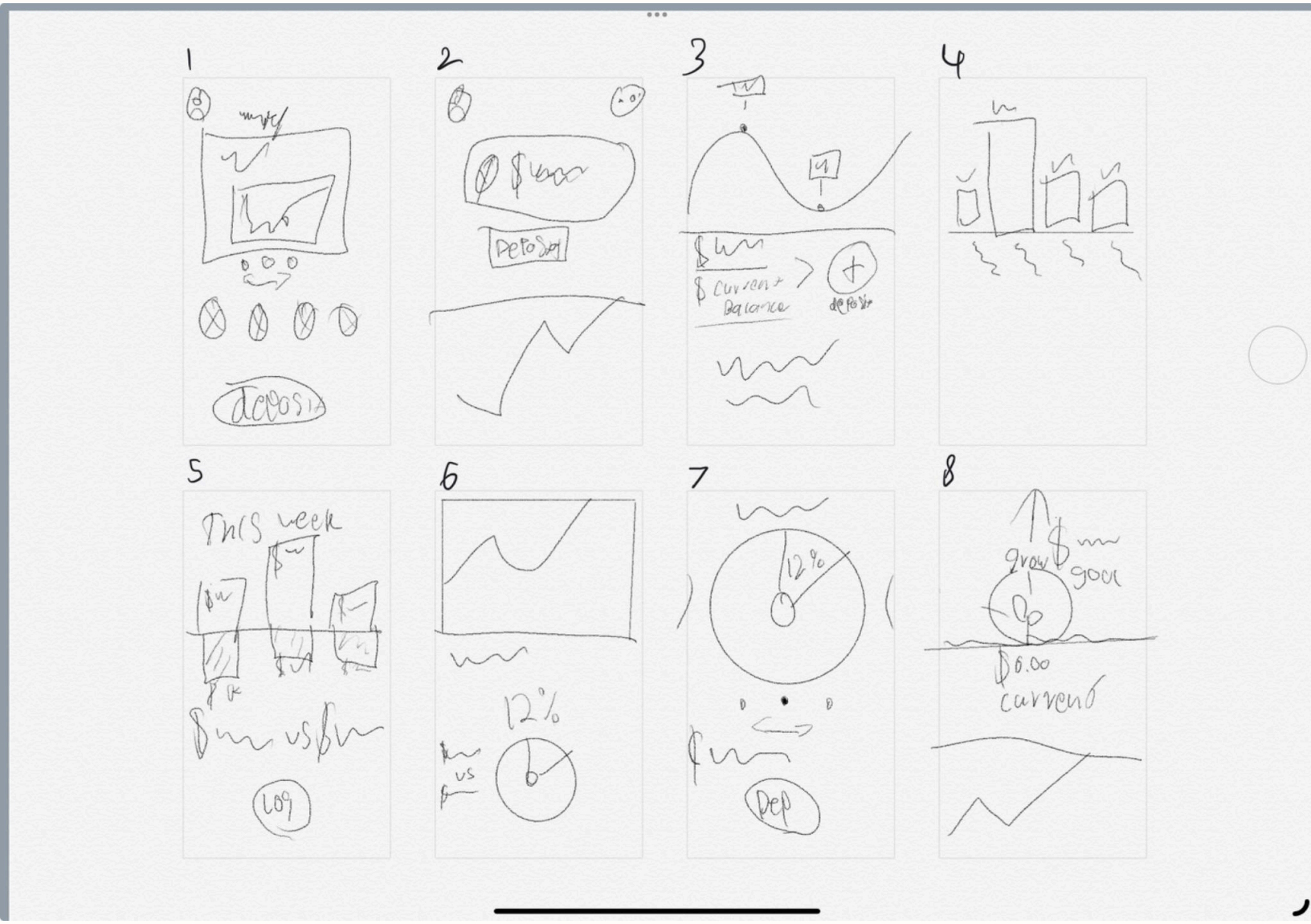
- Dashboard
    - Summary cards for savings goals with progress bars
        - Total savings amount per goal
        - Clickable for detailed goal information
        - Visual summary of goal status
    - Set new goal
    - Recent deposit activity
    - Settings icon
- Goal Setting
    - Inputs for:
        - Goal Name
        - Target Amount
        - Starting Amount
        - Deadline
        - Savings Term (weekly, etc)
- Savings Visualisation
    - Interactive graph
        - Showing savings goal and current savings
    - Easily switching between goals
    - Button to log deposit
- Savings Management (log deposit)
    - Inputs for:
        - Deposit amount
        - Date of deposit record
    - All deposit history
    - Option to change term
- “Detailed Goal Information” View
    - Goal Summary
        - Current savings
        - Target amount
        - Remaining amount to save
    - Graph showing saving progress over time
    - Projection of future savings based on current information
    - % of paycheck needed to put into savings
- Settings
    - Personal Preferences
        - Currency
    - “Link with bank account” option
    - Account settings

Prototype Features

- Goal Setting
    - Interface to easily set new savings goals
    - Option to add a starting amount for each savings goal
    - Input fields for target amount and deadline
- Savings Visualisation
    - Visualisation of the difference between savings and goals
    - Graphs showing savings progress
        - Visual Representation of savings progress
        - Easy access to the graph from the home screen / main dashboard
    - Easy modification of parameters
- Multiple Savings Goals
    - Ability to create and track multiple savings goals simultaneously
    - Separate screens / displays for each goal
- Income Visualisation
    - Visualisation of all expenses in a graph

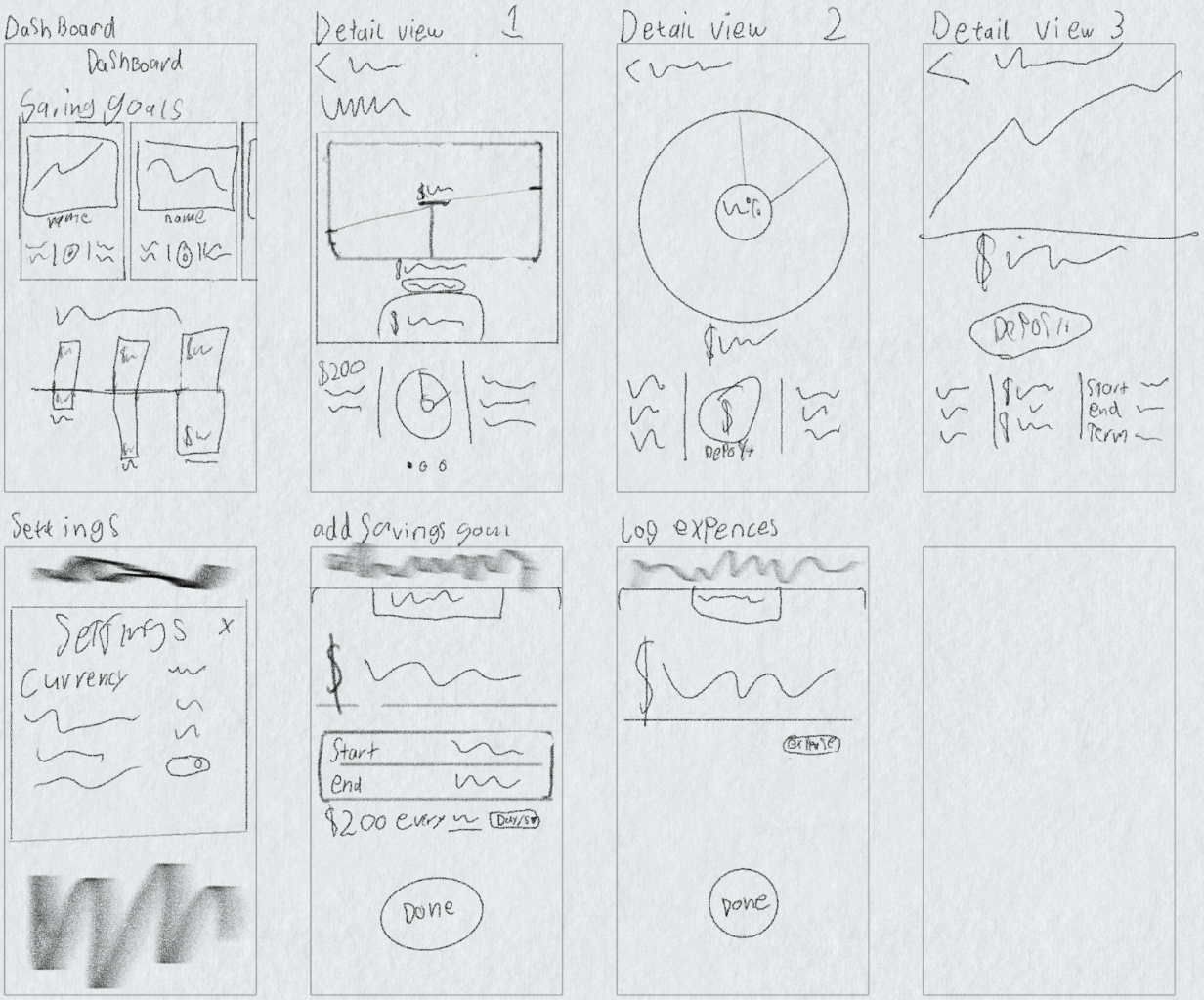
Lo-Fi Prototype

Crazy 8s



I mainly brainstormed graphs / visual representations of data, to ideate on what I will be building in the lo-fi prototype. I found problems with the initial sketches and built on them in the solutions in the lo-fi prototype

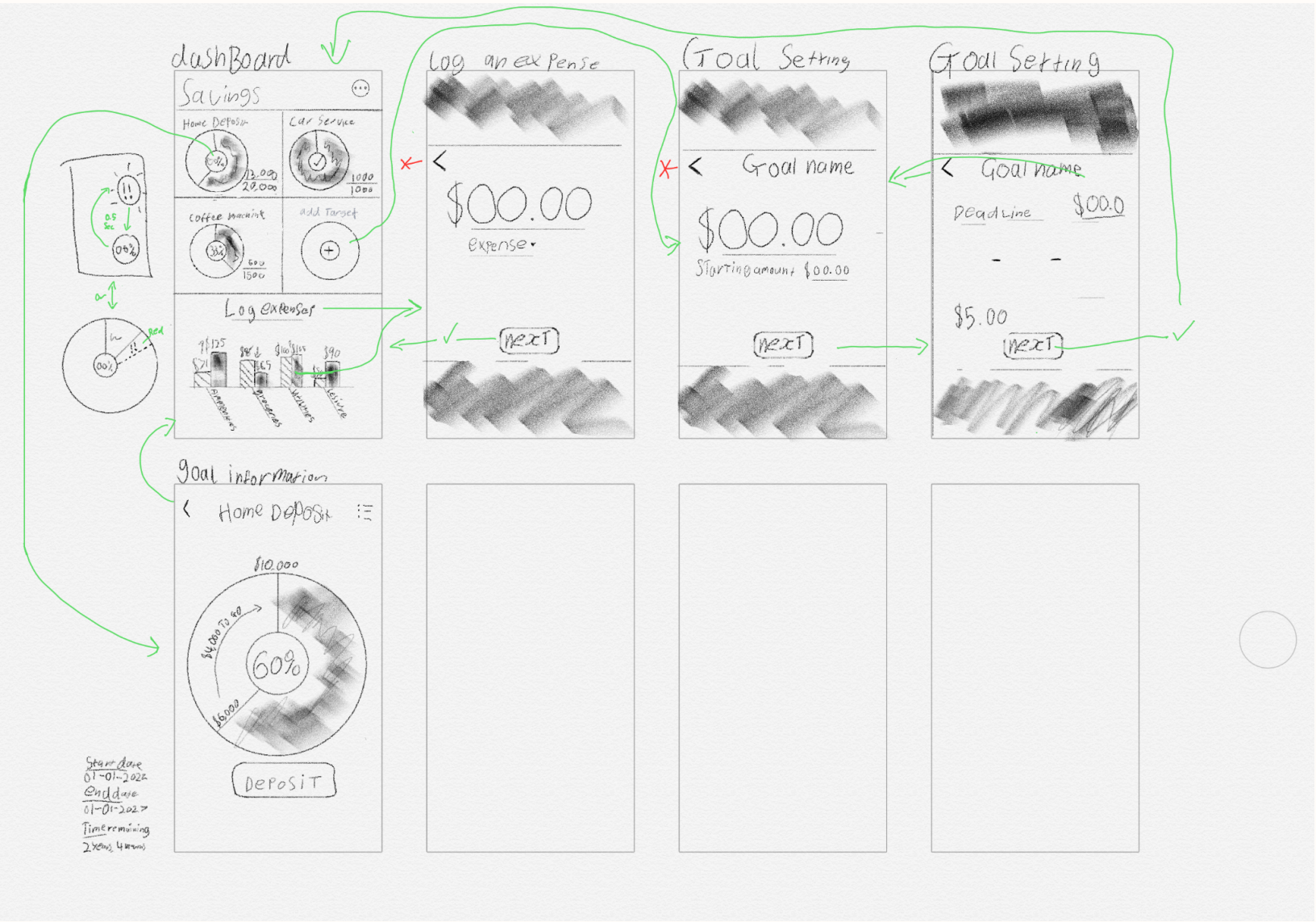
Screens and elements sketching



Lo-fi prototype of all screens that I thought would be relevant in testing, with some variations of different views included to see how they might look in comparison to each other (Detail views).

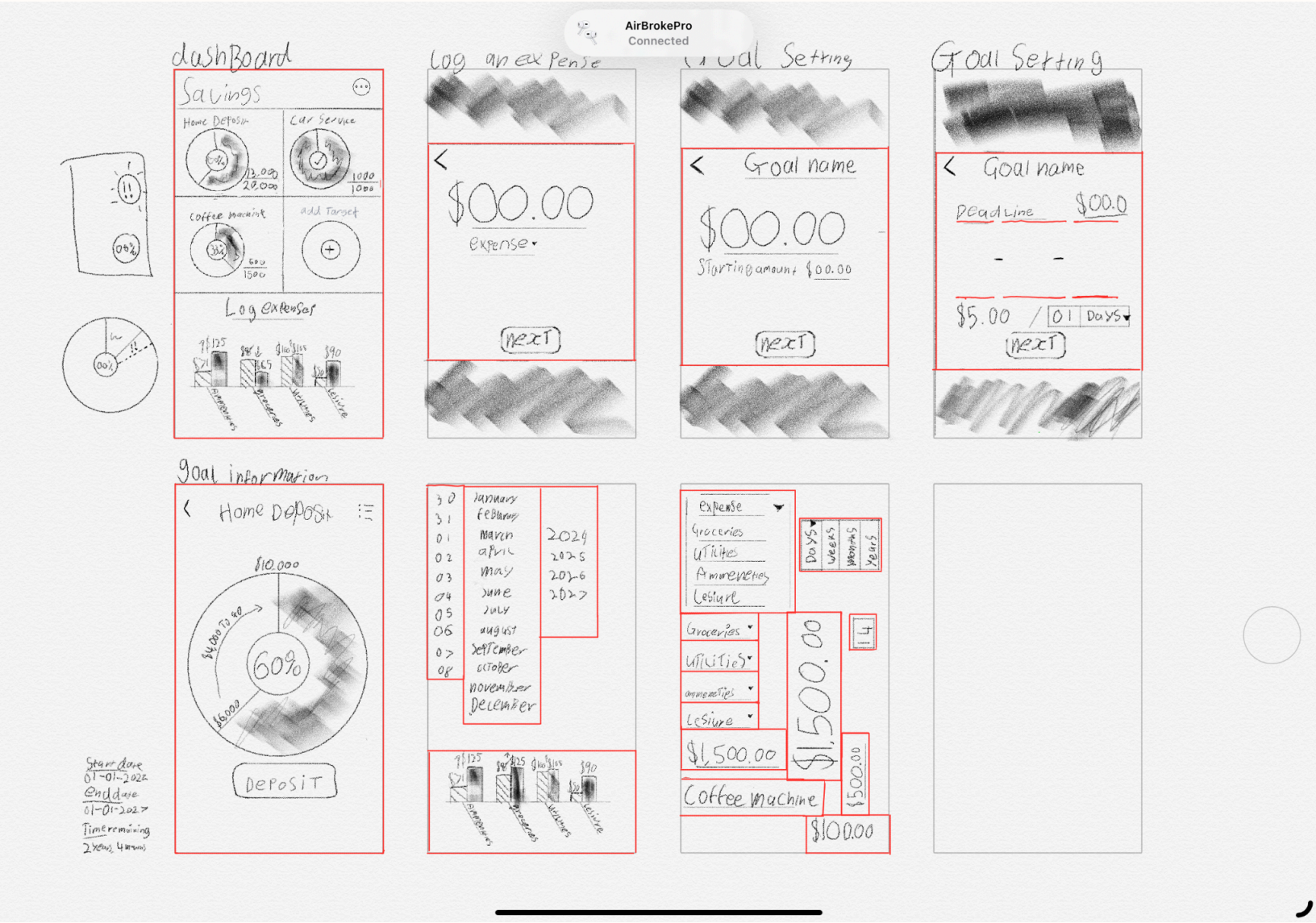
Hi-Fi Prototype

Wireflow



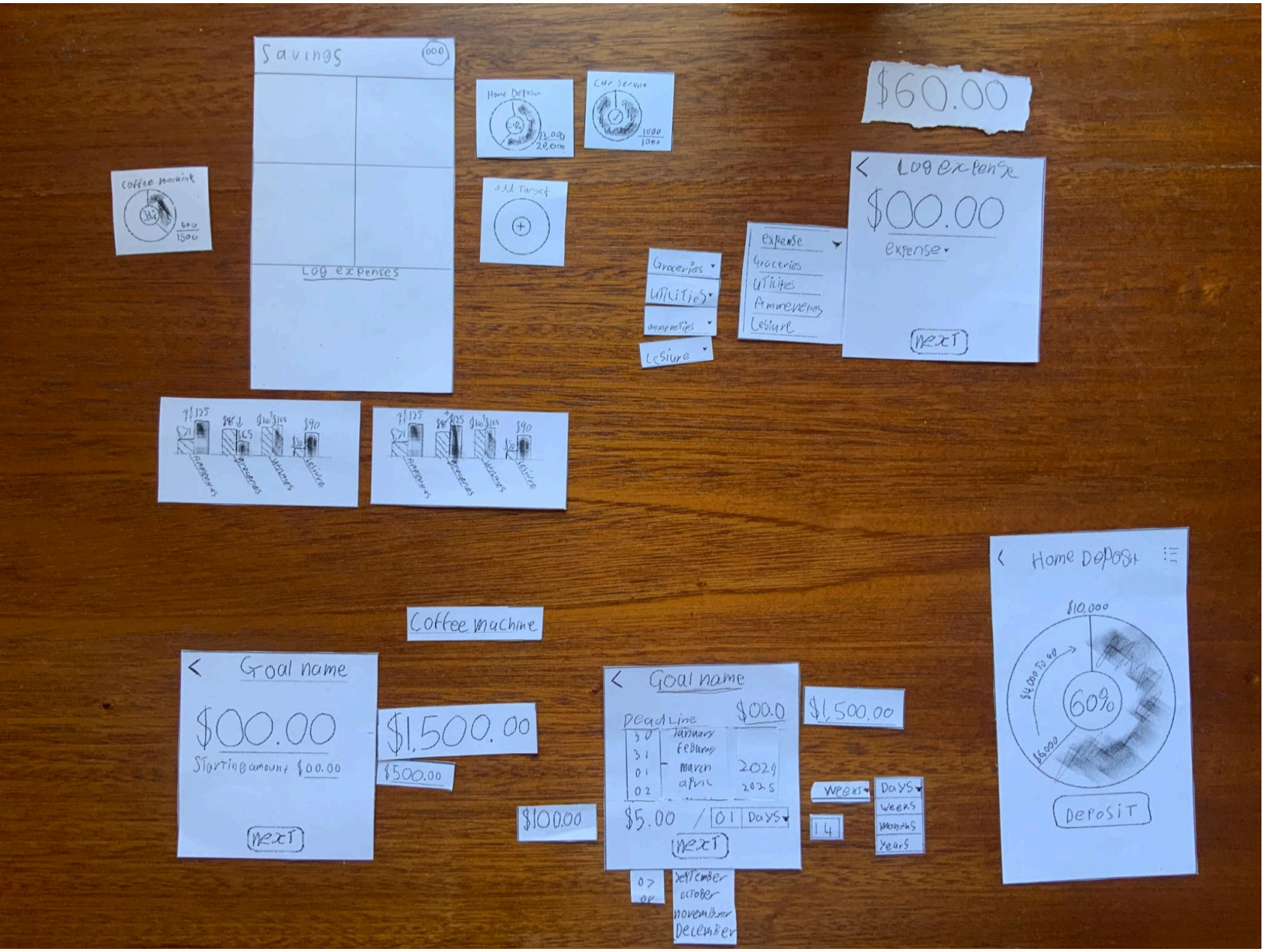
I procedurally sketched and increased the fidelity of my sketches. Unfortunately, I was not able to retrieve or document the process, although I would have liked to comment on why I made certain decisions. Making this on the ipad, although i would later regret it (see user testing) provided me with an environment that made mistakes very easy to fix.

Prototype with element cutouts



The high-fi without the 'flow', including the cutouts I will use when printed out.

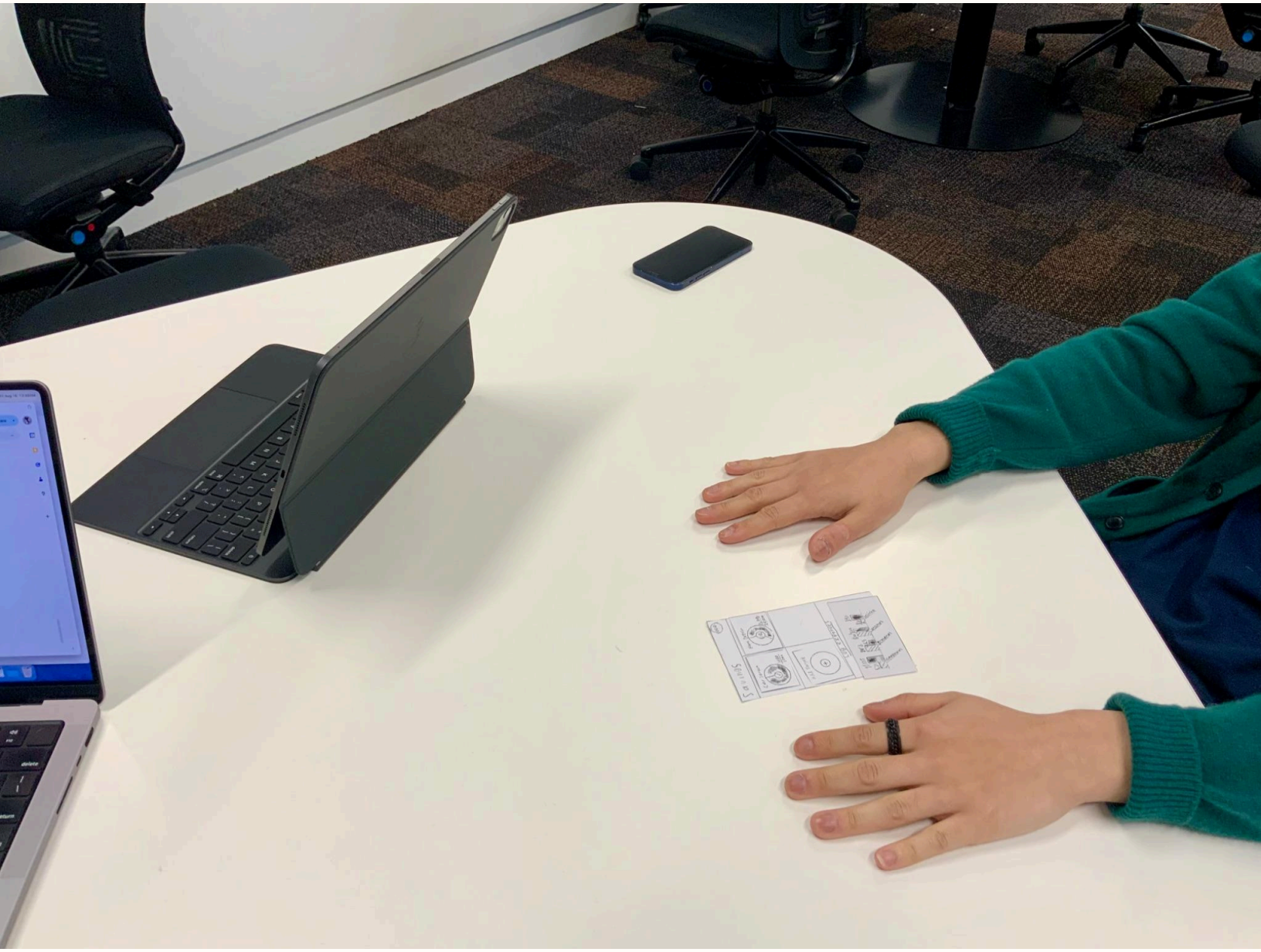
Paper Prototype



Printed out on physical paper, I was able to identify some missing items / elements. I was able to make cutouts on the spot and using the resources I came prepared with (I came with a pencil, I borrowed a ruler) I was able to professionally create the missing elements. I find this to be the most important part of testing on paper, as discussed in the class presentations, They are extremely malleable and easy to edit or make quick changes to.

User Testing

Setup



Scenario

Part 1

You Just bought $60 worth of groceries with cash at queen vic market, and want to keep track of your spendings.

Part 2

On the way out you remembered that you needed to save up some money for a coffee machine for your friend's birthday present.

Sample information you might need to know:

- You are paid every 2 weeks
- You need $1,500 by the end of June, before their birthday
    - You have $500 worth of spare money you can put towards this goal right now.

Calculate how much you need to save per paycheck

Post Test Questionnaire

What did you like?

What didn't you like?

What did you understand?

What didn't you understand?

What would you have done differently?

Did anything happen outside of your expectations, Or need explanation?

Interview 1 - Joshua

https://rmiteduau-my.sharepoint.com/:v:/g/personal/s3972407_student_rmit_edu_au/ERiADYhjI1pIhAvVU0bFqvMBF1ZKtCGjnebkydBM1mVd9w?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=gMhHPX

Observations:

Part 1.

The user had a hard time trying to find the function to log savings

The user was not able to identify or use the function of the category button while logging expenses

Part 2.

Immediately went to add a target with the correct button

Immediately and correctly identified the correct button to add the goal amount

Did not recognise to add a starting amount when presented with the option on the first screen

User immediately chose to set the deadline, and understood the interface.

Had a lot of trouble understanding the information at the bottom of the page
"$5.00 / 01 Days"

Post Test Questions and Answers

Part 1.

- Didn't like the grid layout with the current savings goals, it was a bit too much information
- Couldn't easily identify if the graph was a button or just visual
- Don't really know what 'target' means, felt like it needed further explanation
- Liked the 'Log Expenses' modal. User thought everything was clear, mentioning that you can press on the number to input the amount to log.

Part 2.

- Second modal page was really confusing:
    - Due to the facilitator's (my) lack of attention, information on the second modal was left out (goal amount)
    - The location of the goal amount in line with the word 'deadline' made it seem like they were connected elements

    - Didn’t understand the “$5.00” at the bottom of the screen
    - Didn’t understand the “01 Days” at the bottom of the screen
- User said that the first modal was good and the function to add a saving goal was obvious
-

Interview 2 - Ilyas

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Observations:

Part 1:

- User had a hard time navigating to add an expense
    - Immediately clicked the first button that screamed ‘add’ at them, even though the wording was not related to their goals. Satisficing possibly?
- In the following modals, the user did not input any information or go back, instead navigating completely through the interaction sequence.

- When the user navigated to the desired screen to log expenses, they were able to recognise that they were where they wanted to be..
    - When choosing a category for the expense, the user chose the ‘utilities’ category over the groceries category which would have been correct in this instance. I believe this is due to the poor choice of wording, lack of visual indicators, and the list order.
- The user was unsuccessful in logging their $60 expense

Part 2:

- Due to what i believe to be a lack of clarification on my behalf, the user spent a lot of time navigating through the app giving feedback through this section, which although valuable, was not the desired outcome
- The user navigated through the first modal without inputting any information.
- The user navigated through and finished the modal sequence, only setting the target date for a goal.
    - This might have been due to the interactive sliders, the user may have expected other interactive elements to be the only way to input information
-

Post Test Questions and Answers

Part 1:

- My handwriting was hard to read, specifically “Log Expenses”
    - I must take further time and care to ensure legibility of my handwriting.

Part 2:

- The handwriting was hard for the user to read:
    - Goal Name (savings modal 1)
    - Car Service (saving target on dashboard)
    - Home Deposit (saving target on dashboard)
- The user appreciated the simplicity of the detailed view page.
- The user expected the meatball menu on the home screen to have a function
    - As well as the meatball menu on the detail info screen.
- The information of ‘log expenses’ was hard to read
- Liked the layout, the grid system was organised and consistent
- The user didn’t understand why functions like hamburger menus and deposit button weren’t working
- User suggested not to make the graph too dark
- User would have centre aligned the horizontal position of the title on the dashboard
- User said that the information was ‘well formatted’ regarding the modals
    - This is confusing, because their actions conflict heavily with this statement. Further analysis required.
-

Reflection

Pre-Interview

Using only the iPad app, with no pre-made paper adaptations proved to be almost disastrous, as I had to scramble to get the pages printed and fight technical difficulties, using and going past the allocated class time. Not ideal, to say the least. I would at the very least frame this as a mistake to remember and learn from for future projects.

Post-Interview

I overlooked the keyboards and number-pads as design elements, and forgot to add them into my design, only noticing the lack of them or the expectation that they would be there during user testing.

My design for showing expenses does not currently sufficiently represent a button, users tend to overlook that it is a button as well as information display.

While a unique idea, my deadline scrollers seemed to detract the user's attention from the other components of the prototype. The divider between a couple of the scrollable selectors also broke before I could test the prototypes.

During the second user testing, during a sequence I had not planned for (zero information inputted during setting a saving target) I improvised and allowed the user to navigate through the modals. I believe this was the wrong course of action, and there should have been an indication that the user could continue navigating *after* the user inputted information, or an error message when they attempted to click through.

My modals present too much information per screen, that is easy to be skipped. A more linear approach with less tasks per modal might have been beneficial.

I noticed that while testing, even outside the scope of the scenarios, users would intentionally or unintentionally navigate through every part of my app to see the function. Whether out of curiosity or confusion, the lack of these elements took from the immersion and a large part of some of my feedback was in noticing the lack of these elements.

I could have taken much more care into clarifying my handwriting for certain elements in the prototype.